

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer 2024

Course: B. Tech.

Branch: Mechanical Engineering

Semester: VI

Subject Code & Name: BTMOE605C Energy Conservation and Management

Max Marks: 60

Date: 25/06/2024

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks		
Q.1 Solve Any Two of the following.		12		
A) Explain the concept of Energy Security. What is the importance of energy security for a developing nation like India?	Level 2 Understand CO1	6		
B) Explain the difference between Energy conservation and energy efficiency with suitable examples.	Level 2 Understand CO1	6		
C) Describe the energy conservation opportunities in small and medium scale industries.	Level 2 Understand CO1	6		
Q.2 Solve Any Two of the following.		12		
A) Explain the parameters / observations recorded during Walkthrough Energy Audit.	Level 2 Understand CO1	6		
B) A company wants to set up a reserve which will help the company to have an annual equivalent amount of Rs. 10,00,000 for the next 20 years towards its employee's welfare measures. The reserve is assumed to grow at the rate of 15% annually. Find the single-payment that must be made now as the reserve amount. Also draw cash flow diagram.	Level 3 Apply CO3	6		
C) Explain the parameter recorded using following measuring instruments in energy audit: 1. Stroboscope 2. Power quality analyzer 3. Infrared thermometer 4. Lux meter 5. Ultrasonic flow meter 6. Pitot tube and manometer	Level 2 Understand CO2	6		
Q.3 Solve Any Two of the following.		12		
A) Explain energy conservation techniques in Boiler and Furnace.	Level 2 Understand CO2	6		
B) Consider a project, which has the following cash flow stream. The cost of capital, k, for the firm is 10 %.	Level 3 Apply CO3	6		
<table><tr><td>Investment</td><td>Rs. (10,00,000)</td></tr></table>			Investment	Rs. (10,00,000)
Investment	Rs. (10,00,000)			

Saving in Year	Cash Flow
1	200,000
2	200,000
3	300,000
4	300,000
5	350,000

Calculate the Net Present Value of this project.

- C) In a heat treatment shop, steel components are heat-treated in batches of 80 Tons. The heat treatment cycle is as follows:
- Level 3 6
Apply
CO2
- Increase temperature from 30°C to 850°C in 3 hours.
- Maintain 850°C for 1 hour (soaking time).
- Cool the material to 60°C in 4 hours.
- Calculate the efficiency of the furnace, if the specific heat of steel is 0.12 kcal/kg°C and fuel oil consumption per batch is 1400 litres.
- GCV of fuel oil – 10,200 kcal/kg,
- Cost of fuel oil - Rs. 46,000/kL,
- Sp. gr. of fuel oil - 0.92.
- Quantity of steel treated per batch - 80 Tons.
- Calculate: -
- Mass of fuel consumed
 - Heat given to steel
 - Heat input from fuel
 - Furnace efficiency

Q.4 Solve Any Two of the following.

- A) Explain Topping cycle and Bottoming cycle with block diagram.
- Level 2 12
Understand
CO4 6
- B) What is the purpose of insulators? Enlist name of different insulating material.
- Level 2 6
Understand
CO4
- C) What is the refrigeration load in TR when 15 m³/hr of water is cooled from 21°C to 15°C?
- Level 3 6
Apply
CO4
- If the compressor motor draws 29 kW, chilled water pump draws 4.6 kW, condenser water pump draws 6.1 kW and Cooling Tower fan draws 2.7 kW, what is overall kW/TR?

Q.5 Solve Any Two of the following.

- A) Explain following components of electricity tariff in India.
- Level 3 12
Apply
CO5 6
- MD Charges
 - PF Penalty / incentive
 - TOD Charges
- In an industry, if the drawl over a recording cycle of 30 minutes is:
- 2500 kVA for 4 minutes
- 3600 kVA for 12 minutes
- 4100 kVA for 6 minutes

3800 kVA for 8 minutes

Calculate the Maximum Demand in the plant.

- | | | |
|---|------------------------------|---|
| B) Explain the concept of power factor in electrical system, with its mathematical formula. State the benefits of power factor improvement. | Level 2
Understand
CO5 | 6 |
| C) A chemical industry has kW consumption of 812 kW with demand of 1160 kVA. To improve the power factor & avoid penalty, the company has added 410 kVAR of reactive power at motor end. Calculate | Level 3
Apply
CO5 | 6 |
| <ol style="list-style-type: none">1. Previous Power factor2. Previous Reactive power in the system in kVAR3. Reduced Reactive power in the system in kVAR4. Reduced kVA demand in the system, if kW consumption remains unchanged.5. Improved Power Factor. | | |

***** End *****